

The Sperner property of Artinian complete intersections in characteristic zero

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Abstract

We record the Sperner property for every standard graded Artinian complete intersection over an arbitrary characteristic-zero field: the maximum minimal number of generators of an ideal, including nonhomogeneous ideals, equals the largest coefficient of its complete-intersection Hilbert polynomial. The result is an immediate consequence of OpenAI’s Artinian Eisenbud–Green–Harris theorem and the previously published implication of Harima, Wachi, and Watanabe. We give an explicit account of that implication, eliminate degree-one generators, justify the passage to all ideals by the associated graded filtration, and verify the field scope. No new proof of EGH or new Lefschetz assertion is claimed.

1 Statement and attribution

Let k be a field and let A be a standard graded Artinian k -algebra with $A_0 = k$. Its unique maximal ideal is $\mathfrak{m} = A_{>0}$, and Nakayama’s lemma gives $\mu_A(I) = \dim_k(I/\mathfrak{m}I)$ for every ideal I . The Dilworth number is $\max_{I \triangleleft A} \mu_A(I)$. The algebra has the *Sperner property* when this equals $\max_j \dim_k A_j$. The ideals in this definition include 0, A , and nonhomogeneous ideals.

Theorem 1. *Let k be any characteristic-zero field, let $n \geq 0$, and let*

$$A = k[x_1, \dots, x_n]/(f_1, \dots, f_n), \quad d_i = \deg f_i \geq 1,$$

where the variables have degree one and the f_i form a homogeneous regular sequence. Then

$$\max_{I \triangleleft A} \mu_A(I) = \max_{j \geq 0} \dim_k A_j = \max_{j \geq 0} [t^j] \prod_{i=1}^n (1 + t + \dots + t^{d_i-1}). \quad (1)$$

If p is the last degree attaining the maximum, equality is attained by $\mathfrak{m}^p$. For $n = 0$, or when all $d_i = 1$, the algebra is k , the product is 1, and the maximum is 1, attained by A .

Harima–Wachi–Watanabe [1, Theorem 11] proved that EGH for a given graded complete intersection implies its Sperner property. Their Definition 2 already ranges over all ideals; their Proposition 8 invokes Watanabe’s graded-ideal reduction [2, Lemma 2.4]. OpenAI’s characteristic-zero EGH theorem [3, Corollary 1.3] supplies the previously conditional input. Its companion [4, Theorem 1.1 and Corollary 1.2] proves stronger Betti inequalities, which are not needed here. This note records the resulting consequence and checks its scope; neither the upstream breakthrough nor the known implication and reductions are claimed as new work of this author. The upstream manuscripts are dated September 23, 2026; the public collection announcement is dated October 6, 2026 [5]. No claim of first priority is made.

2 The exact EGH input and scope reductions

The only EGH assertion needed is the following Hilbert-function statement. For $S = k[z_1, \dots, z_q]$ and a full homogeneous regular sequence $g_1, \dots, g_q$ of degrees $a_i \geq 2$, every homogeneous ideal $Q \supseteq (g_1, \dots, g_q)$ has a homogeneous counterpart $Q' \supseteq (z_1^{a_1}, \dots, z_q^{a_q})$ such that

$$\dim_k(S/Q)_j = \dim_k(S/Q')_j \quad \text{for every } j \geq 0. \quad (2)$$

The counterpart may have extra generators in degrees beyond those of Q ; no assertion about its generator count is required. OpenAI [3, Theorem 1.1 and Corollary 1.2] gives this over $\mathbb{C}$ using commuting linear forms with division-algebra coefficients, an ordered basis, and monomial extraction. We invoke that external theorem explicitly, rather than deriving it in this note.

For completeness, its full-length field descent needs no embedding of an arbitrary k into $\mathbb{C}$. For a fixed Q , choose a finitely generated coefficient field $k_0 \subseteq k$ over $\mathbb{Q}$ containing coefficients of generators of Q and of the g_i . Include the g_i in the chosen generators over k_0 . Regularity descends to k_0 by faithful flatness. Embed k_0 into $\mathbb{C}$ and apply the complex theorem after base change. The resulting counterpart can be chosen monomial. Use the same monomial exponents to define it over k_0 and then over k . Dimensions of graded pieces are preserved by both field extensions from k_0 , proving (2) over k . Only full-length EGH is used; no general-length Artinian reduction is needed.

If the original sequence has r linear members, these are linearly independent: dependence would give fewer than n generators for an ideal of height n , contradicting the height theorem. A graded linear coordinate change eliminates them and gives

$$A \simeq k[z_1, \dots, z_{n-r}]/(g_1, \dots, g_{n-r}).$$

The remaining forms retain their degrees, all at least two. Their Artinian quotient makes them a homogeneous system of parameters in a polynomial ring; Cohen–Macaulayness makes them a regular sequence. The graded isomorphism preserves all ideals, the numbers $\mu_A(I)$, and all Hilbert dimensions. Each eliminated factor in (1) is 1. If $r = n$, the case $A = k$ is immediate.

Successive regular-sequence exact sequences give

$$\sum_j h_j t^j = \prod_i (1 + t + \dots + t^{d_i-1}), \quad h_j = \dim_k A_j. \quad (3)$$

Put $c = \sum_i (d_i - 1)$. Artinian complete-intersection duality gives $A_c \simeq k$ and perfect multiplication pairings $A_j \times A_{c-j} \rightarrow A_c$; equivalently the Koszul resolution gives the graded canonical module $A(c)$. The Hilbert function is symmetric and unimodal. An elementary explanation of unimodality is the symmetric chain decomposition used next.

3 Matching and the homogeneous bound

We reproduce the mechanism of [1, Propositions 7–8 and Theorem 11]. Let

$$B = k[z_1, \dots, z_q]/(z_1^{a_1}, \dots, z_q^{a_q}).$$

Its monomial basis is the ranked product of chains $0 \leq e_i \leq a_i - 1$. Such a product admits a symmetric saturated chain decomposition. One direct induction starts with the rectangle $0 \leq i \leq a$, $0 \leq j \leq b$, where $a \leq b$, and uses the chains

$$(r, 0), (r, 1), \dots, (r, b - r), (r + 1, b - r), \dots, (a, b - r), \quad 0 \leq r \leq a.$$

They partition the rectangle and their endpoint ranks sum to $a + b$. Applying this construction to each old chain times a new factor preserves symmetry about half the total rank. Rank sizes are therefore unimodal. Whenever $h_j \leq h_{j+1}$, each chain meeting rank j extends to rank $j + 1$: below the middle this follows from symmetry; at or above the middle no new chain starts, so an ending would force a strict decrease. Chain successors inject any rank- j subset into its upper shadow.

Consequently, for every vector subspace $W \subseteq B_j$,

$$h_j \leq h_{j+1} \implies \dim_k W \leq \dim_k B_1 W. \quad (4)$$

Indeed, echelonize a basis of W by a multiplicative monomial order. Its leading monomials form a set of size $\dim W$. Each surviving variable multiple of a leading monomial is the leading monomial of the corresponding product. Its upper shadow thus lies in the leading monomial set of $B_1 W$, and the chain injection proves the inequality.

For $V \subseteq A_j$, apply (2) to the preimage of the ideal AV in the polynomial ring, obtaining an ideal J of B with $H_{A/AV} = H_{B/J}$. The algebras A, B have the same Hilbert function, and standard grading gives

$$\dim J_j = \dim V, \quad \dim J_{j+1} = \dim A_1 V.$$

For $V = 0$ the conclusion is immediate. Otherwise, if $h_j \leq h_{j+1}$, (4) and the ideal property give

$$\dim V = \dim J_j \leq \dim B_1 J_j \leq \dim J_{j+1} = \dim A_1 V. \quad (5)$$

This is the matching property of A . In particular, Hilbert matching has not been used to assert equality of ideal-generator counts.

Let p be the last maximal Hilbert degree. Matching holds in every degree $j < p$, and symmetry gives $c - p \leq p$. If a homogeneous ideal I has initial degree $\alpha < p$, its truncation $I' = \bigoplus_{j \geq \alpha+1} I_j$ is an ideal and

$$\mu(I') - \mu(I) = \dim A_1 I_\alpha - \dim I_\alpha \geq 0. \quad (6)$$

This follows by counting the degree pieces of $I/\mathfrak{m}I$. Iterating, including the zero-ideal case, yields

$$\mu(I) \leq \mu(I \cap \mathfrak{m}^p). \quad (7)$$

Now assume $I \subseteq \mathfrak{m}^p$, and set $J = \text{Ann}_A(\mathfrak{m}I)$. Gorenstein duality gives $\dim \text{Ann}(U) = \dim A - \dim U$ for every ideal U . To see the equality with the orthogonal complement, compose multiplication with projection onto A_c . A vector orthogonal to the ideal U is orthogonal to every multiple of its elements; nondegeneracy then shows that it annihilates them. Since $\mathfrak{m}J \subseteq \text{Ann}(I)$,

$$\mu(J) \geq \dim \text{Ann}(\mathfrak{m}I) - \dim \text{Ann}(I) = \mu(I). \quad (8)$$

The graded perfect pairings also give $\text{Ann}(\mathfrak{m}^u) = \mathfrak{m}^{c-u+1}$ for $0 \leq u \leq c+1$: one inclusion follows by degree, and any nonzero lower-degree element pairs nontrivially with an element of $\mathfrak{m}^u$. Therefore

$$J \supseteq \text{Ann}(\mathfrak{m}^{p+1}) = \mathfrak{m}^{c-p} \supseteq \mathfrak{m}^p.$$

Applying (7) to J gives $\mu(I) \leq \mu(J) \leq \mu(\mathfrak{m}^p) = h_p$. Applying it first to any homogeneous I proves the homogeneous bound.

4 All ideals and equality

For an arbitrary ideal I , use the finite filtration $F^j A = \mathfrak{m}^j = \bigoplus_{u \geq j} A_u$, and intersection filtrations on I and $\mathfrak{m}I$. Under the natural identification $\text{gr}_{\mathfrak{m}} A \simeq A$, the initial ideal

$$J = \bigoplus_j (I \cap \mathfrak{m}^j) / (I \cap \mathfrak{m}^{j+1})$$

is homogeneous. Its pieces are the lowest-degree initial parts of elements of I . Multiplication of such initial parts shows $\mathfrak{m}J \subseteq \text{gr}_F(\mathfrak{m}I)$. Finite associated grading preserves total dimensions, so

$$\mu(I) = \dim J - \dim \text{gr}_F(\mathfrak{m}I) \leq \dim J - \dim \mathfrak{m}J = \mu(J) \leq h_p.$$

Equality $\text{gr}_F(\mathfrak{m}I) = \mathfrak{m}J$ is neither assumed nor necessary. Finally $\mathfrak{m}^p / \mathfrak{m}^{p+1} \simeq A_p$, so $\mathfrak{m}^p$ attains h_p . This proves Theorem 1.

The generator bound itself also descends along a field extension K/k , if one is used: flatness gives

$$(IA_K) / (\mathfrak{m}_K IA_K) \simeq (I / \mathfrak{m}I) \otimes_k K, \quad \mu_{A_K}(IA_K) = \mu_A(I),$$

and $\dim_K(A_K)_j = h_j$. An upper bound over K descends to every ideal over k ; the equality witness already exists over k . No descent of arbitrary ideals from K is asserted.

5 Examples and limitations

The following Hilbert functions are coefficient lists in increasing degree. They illustrate (1), rather than serve as evidence for EGH.

degrees	Hilbert function	maximum
$\emptyset$ or $(1, 1, 1)$	1	1
$(1, 2, 4)$	1, 2, 2, 2, 1	2
$(2, 2, 2)$	1, 3, 3, 1	3
$(2, 3, 4)$	1, 3, 5, 6, 5, 3, 1	6
$(3, 3, 3)$	1, 3, 6, 7, 6, 3, 1	7
eight quadrics	1, 8, 28, 56, 70, 56, 28, 8, 1	70

For n quadrics the maximum is $\binom{n}{\lfloor n/2 \rfloor}$. For instance, in the eight-quadric case every ideal has at most 70 generators and $\mathfrak{m}^4$ has exactly 70. The accompanying integer program reproduces the examples by convolution and independent enumeration of bounded exponent vectors.

The theorem concerns characteristic zero and standard graded Artinian complete intersections. It supplies no weak or strong Lefschetz property, no assertion for nongraded complete intersections, and no unrestricted positive-characteristic assertion. The downstream conditional argument and monomial matching work over any field; the restriction comes from the general EGH input used here.

Disclosure and verification. AI tools were used extensively in research, drafting, computation and verification. The mathematical input from OpenAI is itself attributed to OpenAI using the supplied manuscript citations. Internal automated adversarial audits are documented in the supplement; neither they nor the calculations constitute conventional human peer review. This preprint has not undergone conventional human peer review or refereeing at publication. No formalization of this follow-on theorem is claimed. The source/dependency hashes and reproduction instructions accompany it.

References

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